EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Oxidation and Reduction

1. Consider the redox reaction $2S_2O_3^{2\ominus} + I_2 \longrightarrow S_4O_6^{2\ominus} + 2I^{\ominus}$:

(A) $2S_2O_3^{2\ominus}$ gets oxidised to $S_4O_6^{2\ominus}$ (B) I_2 gets reduced to $I^{\ominus}$

(C) I_2 gets oxidised to $I^{\ominus}$ (D) Both (A) & (B)

CRX096

2. The reaction $H_2S + H_2O_2 \rightarrow S + 2H_2O$ shows :

(A) Oxidation of H_2O_2 (B) Reduction of H_2O_2

(C) Acidic nature of H_2O_2 (D) Alkaline nature of H_2O_2

CRX086

3. If in a chemical reaction, $Fe^{2\oplus}$ is converted into $Fe^{3\oplus}$, then $Fe^{2\oplus}$:

(A) Gains one electron and gets reduced

(B) Looses one electron and gets oxidized

(C) Gains one proton and gets oxidised

(D) Looses one proton and gets reduced

CRX087

4. In the reaction, $ZnO + C \rightarrow Zn + CO$:

(A) ZnO is reduced (B) C is oxidised

(C) Above reaction is a redox reaction (D) All of above

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CRX101

5. $CH_4\,(g) + 2O_2\,(g) \rightarrow CO_2\,(g) + 2H_2O\,(g)$

In above reaction :

(A) both CH_4 and O_2 are oxidised (B) both CH_4 and O_2 are reduced

(C) CH_4 is oxidised and O_2 is reduced (D) CH_4 is reduced and O_2 is oxidized

CRX102

6. In the reaction, $Zn + CuCl_2 \rightarrow ZnCl_2 + Cu$:

(A) Zn is getting reduced (B) $CuCl_2$ is getting oxidized

(C) Both (A) and (B) are correct (D) None of (A) and (B) are correct

CRX103

7. In the reaction, $Mg + F_2 \rightarrow MgF_2$, correct statement(s) is/are :

(A) Both oxidation and reduction take place in above reaction

(B) Mg is oxidized

(C) F_2 is Reduced

(D) All of the above

CRX104

8. In the reaction, $3Mg + N_2 \rightarrow Mg_3N_2$, incorrect statement is :

(A) Mg is reduced (B) Mg is oxidised

(C) N_2 is oxidised (D) Both (A) and (C)

CRX105

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9. $FeCl_2 + \frac{1}{2}Cl_2 \rightarrow FeCl_3$

In above reaction :

(A) Fe is oxidised
(B) Cl_2 is reduced
(C) both (A) and (B) are true
(D) both (A) and (B) are false

CRX106

Oxidation Number

10. Which statement is incorrect :

(A) Oxidation number of oxygen is –2 in most of its compounds.
(B) Oxidation number of oxygen is +2 in Oxygen difluoride.
(C) Oxidation number of oxygen is $-\frac{1}{2}$ in superoxides.
(D) Oxidation number of oxygen is +1 in peroxides.

CRX088

11. Identify the correct statement :

(A) Halogens always have –1 oxidation state in their compounds.
(B) Oxidation number can be zero, integer or fractional.
(C) In OF_2, the oxidation number of F is +1.
(D) Hydrogen always has + 1 oxidation number in its compounds.

CRX089

12. Among the following compounds given below, what is the sum of the oxidation states of all underlined elements?

$\underline{C}O_2$, $K_2\underline{Mn}O_4$

(A) 8 (B) 9 (C) 10 (D) 12

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CRX090

13. The oxidation number of Phosphorus in $Mg_2P_2O_7$ is :

(A) + 3 (B) + 2 (C) + 5 (D) – 3

CRX091

14. The oxidation number of sulphur in S_8 & S_2F_2 respectively are :

(A) 0, +1 (B) +2, +1 (C) 0, –1 (D) –2, +1

CRX092

15. Manganese achieves its maximum oxidation state in which of these compounds :

(A) MnO_2 (B) Mn_3O_4 (C) $KMnO_4$ (D) K_2MnO_4

CRX093

16. The incorrect order of decreasing oxidation number of S in compounds is :

(A) $H_2S_2O_7 > Na_2S_4O_6 > Na_2S_2O_3 > S_8$
(B) $H_2SO_5 > H_2SO_3 > SCl_2 > H_2S$
(C) $SO_3 > SO_2 > S_8 > H_2S$
(D) $H_2SO_4 > SO_2 > H_2S > H_2S_2O_8$

CRX109

17. Identify the oxidant in the following reaction :

$KMnO_4 + KCl + H_2SO_4 \longrightarrow MnSO_4 + K_2SO_4 + H_2O + Cl_2$

(A) $KMnO_4$ (B) KCl (C) H_2SO_4 (D) H_2O

CRX110

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18. Identify the reductant in the following reaction :

$FeCl_2 + H_2O_2 + HCl \longrightarrow FeCl_3 + H_2O$

(A) $FeCl_2$ (B) H_2O_2 (C) HCl (D) $Cl^{\ominus}$

CRX111

19. H_2O_2 acts as a oxidising agent in :

(A) $FeCl_2 + HCl + H_2O_2 \longrightarrow FeCl_3 + H_2O$ (B) $HI + H_2O_2 \longrightarrow I_2 + H_2O$

(C) $H_2SO_3 + H_2O_2 \longrightarrow H_2SO_4 + H_2O$ (D) All of these

CRX112

20. (A) $S^{2\ominus} + 4H_2O_2 \longrightarrow SO_4^{2\ominus} + 4H_2O$

(B) $Cl_2 + H_2O_2 \longrightarrow 2HCl + O_2$

The true statement regarding the above reactions is :

(A) H_2O_2 acts as reductant in both the reactions.

(B) H_2O_2 acts as oxidant in reaction (A) and reductant in reaction (B).

(C) H_2O_2 acts as an oxidant in both the reactions.

(D) H_2O_2 acts as reductant in reaction (A) and oxidant in reaction (B).

CRX113

Types of Redox Reactions

21. Which of the following is/are decomposition reaction(s) ?

(A) $2HgO \longrightarrow 2Hg + O_2$ (B) $Fe_2(SO_4)_3 \longrightarrow Fe_2O_3 + SO_3$

(C) $2H_2O \longrightarrow 2H_2 + O_2$ (D) All of these

CRX097

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22. Which is a displacement redox reaction :

(A) $Cr_2(SO_4)_3 + 6KOH \rightarrow 2Cr(OH)_3 + 3K_2SO_4$ (B) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

(C) $NH_4Cl + NaOH \rightarrow NH_3 + NaCl + H_2O$ (D) $2AgNO_3 + Cu \rightarrow Cu(NO_3)_2 + Ag$

CRX098

23. Which of the following is redox reaction?

(A) $NaOH + HCl \longrightarrow NaCl + H_2O$ (B) $AgNO_3 + KI \longrightarrow AgI + KNO_3$

(C) $BaO_2 + H_2SO_4 \longrightarrow BaSO_4 + H_2O_2$ (D) $SnCl_2 + HgCl_2 \longrightarrow SnCl_4 + Hg$

CRX114

24. Which of the following is a combination redox reaction ?

(A) $CaO + CO_2 \longrightarrow CaCO_3$ (B) $H_2 + Cl_2 \longrightarrow 2HCl$

(C) $CaO + 2HCl \longrightarrow CaCl_2 + H_2O$ (D) $NaOH + HCl \longrightarrow NaCl + H_2O$

CRX115

Disproportionation, Comproportionation, Intramolecular Redox

25. Which reaction does not represent auto redox or disproportionation :

(A) $Cl_2 + OH^{\ominus} \longrightarrow Cl^{\ominus} + ClO_3^{\ominus} + H_2O$ (B) $2H_2O_2 \longrightarrow H_2O + O_2$

(C) $2Cu^{\oplus} \longrightarrow Cu^{2\oplus} + Cu$ (D) $(NH_4)_2Cr_2O_7 \longrightarrow N_2 + Cr_2O_3 + 4H_2O$

CRX099

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26. Which of the following is not a comproportionation redox reaction?

(A) $Na_2SO_3 + S \rightarrow Na_2S_2O_3$ (B) $2BaO + O_2 \rightarrow 2BaO_2$

(C) $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$ (D) $SO_2 + 2H_2S \rightarrow 2H_2O + 3S$

CRX100

27. Which of the following is example of disproportionation reaction :

(A) $HgO \longrightarrow Hg + O_2$ (B) $KClO_3 \longrightarrow KCl + O_2$

(C) $KClO_3 \longrightarrow KClO_4 + KCl$ (D) $AgNO_3 + KI \longrightarrow AgI + KNO_3$

CRX116

Equivalent weight & n-factor

28. The equivalent weight of metal (Atomic Mass = 24 u), if 40 g metal oxide contains 60% metal by mass, is :

(A) 8 (B) 12 (C) 16 (D) 24

CRX118

29. Equivalent mass of H_3PO_4 in below reaction is :

$H_3PO_4 + 2\ NaOH \longrightarrow Na_2HPO_4 + 2H_2O$

(A) 98 (B) 49 (C) 32.67 (D) 196

CRX119

30. In acidic medium, potassium dichromate acts as an oxidant according to the equation :

$Cr_2O_7^{2\ominus} + 14H^{\oplus} + 6e^{\ominus} \longrightarrow 2Cr^{3\oplus} + 7H_2O$

What is the equivalent weight of $K_2Cr_2O_7$ (mol. wt.= M) ?

(A) M (B) M/2 (C) M/3 (D) M/6

CRX120

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31. The equivalent weight of $Na_2S_2O_3$ (Mol. wt. = M) in the reaction, $2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$ is :

(A) M/4 (B) M/3 (C) M/2 (D) M

CRX121

32. Equivalent mass of oxidising agent in the reaction, $SO_2 + 2H_2S \longrightarrow 3S + 2H_2O$ is :

(A) 32 (B) 64 (C) 16 (D) 8

CRX122

33. In the change, $3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$

if the atomic weight of iron is 56 u, then its equivalent weight will be :

(A) 42 (B) 21 (C) 63 (D) 84

CRX123

34. Equivalent weight of FeC_2O_4 in the change : $FeC_2O_4 \longrightarrow Fe^{3\oplus} + CO_2$ is : (mol. wt.= M)

(A) M/3 (B) M/6 (C) M/2 (D) M/1

CRX124

35. The valence factor of Cu_2S in below change is :

$Cu_2S \longrightarrow Cu^{2\oplus} + SO_4^{2\ominus}$

(A) 2 (B) 4 (C) 8 (D) 10

CRX125

36. In the reaction, $VO + Fe_2O_3 \rightarrow FeO + V_2O_5$, the eq. wt. of V_2O_5 is equal to its :

(A) Mol. wt. (B) Mol. wt./8 (C) Mol.wt./6 (D) Mol. wt./2

CRX128

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37. In the reaction : $A^{n_2\ominus} + xe^{\ominus} \longrightarrow A^{n_1\ominus}$, here x will be

(A) $n_1 + n_2$ (B) $n_2 - n_1$ (C) $n_1 - n_2$ (D) $n_1 . n_2$

CRX129

38. In a reaction 4 mole of electrons are transferred to one mole of HNO_3 when it acts as an oxidant. The possible reduction product is :

(A) (1/2) mole N_2 (B) (1/2) mole N_2O (C) 1 mole of NO_2 (D) 1 mole NH_3

CRX130

39. The equivalent weight of $MnSO_4$ is half of its molecular weight when it is converted to :-

(A) Mn_2O_3 (B) MnO_2 (C) $MnO_4^{\ominus}$ (D) $MnO_4^{2\ominus}$

CRX131

40. $Cr_2O_7^{2\ominus} + I^{\ominus} + H^{\oplus} \rightarrow Cr^{3\oplus} + I_2 + H_2O$

The equivalent weight of the reductant in the above equation is :–(At. wt. of Cr = 52, I = 127)

(A) 26 (B) 127 (C) 63.5 (D) 10.4

CRX132

Balancing of Redox Reaction

41. In the reaction, $xHI + yHNO_3 \longrightarrow NO + I_2 + H_2O$:

(A) x = 3, y = 2 (B) x = 2, y = 3 (C) x = 6, y = 2 (D) x = 6, y = 1

CRX251

42. For the redox reaction, $Cr_2O_7^{2\ominus} + C_2O_4^{2\ominus} + H^{\oplus} \longrightarrow Cr^{3\oplus} + CO_2 + H_2O$,

the correct whole number stoichiometric coefficients of $Cr_2O_7^{2\ominus}$, $C_2O_4^{2\ominus}$ and $H^{\oplus}$ are respectively :

(A) 1, 3, 14 (B) 14, 3, 1 (C) 3, 14, 1 (D) 1, 14, 3

CRX252

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43. For the redox reaction $xP_4 + yHNO_3 \longrightarrow H_3PO_4 + NO_2 + H_2O$, upon balancing with whole number coefficients :

(A) x = 1, y = 5 (B) x = 2, y = 10 (C) x = 1, y = 20 (D) x = 1, y = 15

CRX253

44. $xNO_3^{\ominus} + yI^{\ominus} + zH^{\oplus} \rightarrow 2NO + 3I_2 + 4H_2O$; x, y, z respectively in the above equation are :

(A) 2, 6, 8 (B) 1, 6, 4 (C) 0, 6, 8 (D) 2, 3, 4

CRX254

45. In the reaction $X^{\ominus} + XO_3^{\ominus} + H^{\oplus} \longrightarrow X_2 + H_2O$, the molar ratio in which $X^{\ominus}$ and $XO_3^{\ominus}$ react is :

(A) 1 : 5 (B) 5 : 1 (C) 2 : 3 (D) 3 : 2

CRX255

46. Balance the following equation and choose the quantity which is the sum of the coefficients of reactants and products :

..... $KMnO_4$ +..... H_2O_2 +..... H_2SO_4 ..... $\longrightarrow MnSO_4$ +..... O_2 +..... H_2O +..... K_2SO_4

(A) 26 (B) 23 (C) 28 (D) 22

CRX257

47. $CN^{\ominus}$ is oxidised by $NO_3^{\ominus}$ in presence of acid :

$a\,CN^{\ominus} + b\,NO_3^{\ominus} + c\,H^{\oplus} \longrightarrow (a+b)\,NO + a\,CO_2 + \frac{C}{2}H_2O$

What are the whole number values of a, b, c in that order :

(A) 3, 7, 7 (B) 3, 10, 7 (C) 3, 10, 10 (D) 3, 7, 10

CRX259

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48. The number of electrons to balance the following equation, $NO_3^{\ominus} + 4H^{\oplus} + e^{\ominus} \rightarrow 2H_2O + NO$ is :

(A) 5 (B) 4 (C) 3 (D) 2

CRX249

49. Number of moles of electrons taken up when 1 mole of $NO_3^{\ominus}$ ions is reduced to 1 mole of NH_2OH is

(A) 2 (B) 4 (C) 5 (D) 6

CRX250

50. When arsenic sulphide is boiled with NaOH, sodium arsenite and sodium thioarsenite are formed according to reaction :

$xAs_2S_3 + y\ NaOH \longrightarrow xNa_3AsO_3 + xNa_3AsS_3 + \frac{y}{2} H_2O$. What are the values of x and y?

(A) 1, 6 (B) 2, 8 (C) 2, 6 (D) 1, 4

CRX258

Equivalent Concept

51. The mass of oxalic acid crystals ($H_2C_2O_4.2H_2O$) that would contain it's 0.01 equivalents is :

(A) 4.5 g (B) 6.3 g (C) 0.63 g (D) 0.45 g

CRX127

52. The number of milliequivalents in 1.58 g of $KMnO_4$ in a reaction where it gets reduced to $Mn^{2\oplus}$ is :(At. wt. : K = 39 u, Mn = 55 u)

(A) 50 (B) 2 (C) 5 (D) 10

CRX192

53. What volume in litre of 0.1 N $K_2Cr_2O_7$ solution should be taken to have 1 equivalent of $K_2Cr_2O_7$ in a reaction where it is reduced to $Cr^{3\oplus}$?

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(A) 1 L (B) 0.1 L (C) 100 L (D) 10 L

CRX193

54. The milliequivalents of H_3PO_4 in 120 mL of 1.5 M H_3PO_4 solution in a reaction where it undergoes complete neutralization is :

(A) 180 (B) 360 (C) 540 (D) 60

CRX194

55. In the redox reaction, $MnO_4^{\ominus} + C_2O_4^{2\ominus} + H^{\oplus} \longrightarrow Mn^{2\oplus} + CO_2 + H_2O$, 20 mL of 0.1 M $KMnO_4$ react quantitatively with :

(A) 20 mL of 0.1 M oxalate (B) 40 mL of 0.1 M oxalate

(C) 50 mL of 0.25 M oxalate (D) 50 mL of 0.1 M oxalate

CRX195

56. One mole of acidified $K_2Cr_2O_7$ on reaction with sufficient KI will liberate ..... mole(s) of I_2, along with $Cr^{3\oplus}$.

(A) 6 (B) 1 (C) 7 (D) 3

CRX196

57. 1 mol each of H_3PO_2, H_3PO_3 and H_3PO_4 will neutralise respectively x mol of NaOH, y mol of $Ca(OH)_2$ and z mol of $Al(OH)_3$ (assuming all as strong electrolytes). x, y, z are in the ratio :

(A) 3 : 1.5 : 1 (B) 1 : 2 : 3 (C) 3 : 2 : 1 (D) 1 : 1 : 1

CRX199

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58. 125 mL of 63% (w/v) $H_2C_2O_4.2H_2O$ solution is made to react with 125 mL of a 40%(w/v) NaOH solution. The resulting solution is: (ignoring hydrolysis of ions)
(A) neutral (B) acidic (C) strongly acidic (D) alkaline

CRX200

59. A substance which participates readily in both acid-base and oxidation-reduction reactions is :
(A) Na_2CO_3 (B) KOH (C) $KMnO_4$ (D) $H_2C_2O_4$

CRX201

60. What mass of MnO_2 is reduced by 35 ml of 0.16 N oxalic acid ($H_2C_2O_4$) in acidic solution? The skeleton equation is :-

$MnO_2 + H^{\oplus} + H_2C_2O_4 \rightarrow CO_2 + H_2O + Mn^{2\ominus}$

(A) 8.7 g (B) 0.24 g (C) 0.84 g (D) 43.5 g

CRX203

Redox Titration

61. Which of the following solutions will exactly oxidize 25 mL of an acidic solution of 0.1 M iron (II) oxalate:
(A) 25 mL of 0.1 M $K_2Cr_2O_7$ (B) 25 mL of 0.2 M $K_2Cr_2O_7$
(C) 25 mL of 0.6 M $K_2Cr_2O_7$ (D) 12.5 mL of 0.1 M $K_2Cr_2O_7$

CRX224

62. How many litres of Cl_2 gas at NTP (1 ATM & 273K) will be liberated by the oxidation of NaCl with 15.8 g $KMnO_4$ in acidic medium ? (Atomic weight : Mn = 55 u and K = 39 u)
(A) 5.6 (B) 11.2 (C) 56 (D) 1.12

CRX225

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63. 1.04 g of Na_3AsO_4 is boiled with excess of solid KI in presence of strong HCl. The iodine evolved is absorbed in KI solution and titrated against 0.2 N hypo solution. Assuming the reaction to be

$AsO_4^{3\ominus} + 2H^{\oplus} + 2I^{\ominus} \longrightarrow AsO_3^{3\ominus} + H_2O + I_2$

calculate the volume of hypo consumed. [Atomic weight of As = 75 u]
(A) 100 mL (B) 50 mL (C) 25 mL (D) 5 mL

CRX226

64. Dichloroacetic acid ($CHCl_2COOH$) is oxidised to CO_2, H_2O and Cl_2 by 600 m.eq. of an oxidising agent. Same amount of acid can neutralize how many moles of ammonia to form ammonium dichloroacetate :
(A) 0.0167 (B) 0.1 (C) 0.3 (D) 0.6

CRX227

65. If 9.1 g of V_2O_5 is dissolved in acid and is reduced to $V^{2\oplus}$ by zinc metal, how many mole of I_2 could be reduced by the resulting solution, if it is further oxidised to $VO^{2\oplus}$ ions :
[Assume no change in state of $Zn^{2\oplus}$ ions] (Atomic masses : V = 51 u, O = 16 u, I = 127 u)
(A) 0.1 (B) 0.2 (C) 0.05 (D) 0.4

CRX228

66. How many gram of $KMnO_4$ are contained in 4 litre of 0.05 N solution. The $KMnO_4$ is to be used as an oxidant in acidic medium ?
(A) 1.58 g (B) 15.8 g (C) 6.32 g (D) 31.6 g

CRX236

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67. 40 ml, 0.5 M - $Na_2S_2O_3$ solution is required for complete reaction with 25 ml I_2 solution. The molarity of I_2 solution is -

(A) 0.4M (B) 0.8 M (C) 0.25 M (D) 0.16 M

CRX237

68. 25 ml acidified of $K_2Cr_2O_7$ solution is diluted to 200 ml. If 40 ml of the diluted solution exactly required 15 ml of 0.2 N - KI solution for titration, the normality of original $K_2Cr_2O_7$ solution is -

(A) 0.3 N (B) 0.2 N (C) 0.6 N (D) 0.1 N

CRX239

Acid Base Titration

69. 20 ml of 0.8 M - H_2SO_4 solution is diluted to 100 ml. 25 ml of the diluted solution exactly require 30 ml NaOH solution for titration. The molarity of NaOH solution is -

(A) $\frac{2}{15}$M (B) $\frac{4}{15}$M (C) $\frac{1}{15}$M (D) $\frac{1}{60}$M

CRX238

Back Titration

70. 10 g $CaCO_3$ were dissolved in 250 mL of 1 M HCl. What volume of 2 M KOH would be required to neutralise excess HCl ?

(A) 75 mL (B) 25 mL (C) 15 mL (D) 50 mL

CRX202

71. 35 ml of 1.5 M – H_2SO_4 solution is mixed with 50 ml of 0.6 M-NaOH solution. If the resulting solution exactly requires 75 ml of KOH solution for the neutralisation of excess acid, the molarity of KOH solution is -

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(A) 0.5M (B) 0.75 M (C) 2.0 M (D) 1.0 M

CRX240

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Oxidation Number

1. In the compound $NOClO_4$, the oxidation state of

(A) nitrogen is +1 (B) nitrogen is +3 (C) chlorine is +5 (D) chlorine is +7

CRX364

2. Which of the following is/are peroxide(s)?

(A) PbO_2 (B) H_2O_2 (C) SrO_2 (D) BaO_2

CRX365

3. Which of the following is a non-redox process?

(A) $SO_4^{2-} \rightarrow SO_3$ (B) $Cr_2O_7^{2-} \rightarrow CrO_4^{2-}$ (C) $PO_4^{3-} \rightarrow P_2O_7^{4-}$ (D) $C_2O_4^{2-} \rightarrow CO_2$

CRX366

Balancing of Redox Reaction

4. In the reaction, $xZn + yHNO_3(dil) \longrightarrow aZn(NO_3)_2 + bH_2O + cNH_4NO_3$, upon balancing with simplest whole number coefficients, select the correct option(s) :

(A) The sum of the coefficients (a + b + c) is 8.

(B) Whole of HNO_3 is reduced to $NH_4^{\oplus}$ in above reaction.

(C) Zn acts as reductant.

(D) HNO_3 acts as oxidant.

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CRX260

5. For the reaction : $I^{\ominus} + ClO_3^{\ominus} + H_2SO_4 \longrightarrow Cl^{\ominus} + HSO_4^{\ominus} + I_2$

The correct statement(s) in the balanced equation is/are :

(A) Stoichiometric coefficient of $HSO_4^{\ominus}$ is 6. (B) Iodide is oxidized.

(C) Sulphur is reduced. (D) H_2O is one of the products.

CRX117

Equivalent Concept

6. Which of the following samples of reducing agents is/are chemically equivalent to 25 mL of 0.2 N $KMnO_4$ to be reduced to $Mn^{2\oplus}$ and water :

(A) 25 mL of 0.2 M $FeSO_4$ to be oxidized to $Fe^{3\oplus}$

(B) 50 mL of 0.1 M H_3AsO_3 to be oxidized to H_3AsO_4

(C) 25 mL of 0.1 M H_2O_2 to be oxidized to $H^{\oplus}$ and O_2

(D) 25 mL of 0.1 M $SnCl_2$ to be oxidized to $Sn^{4\oplus}$

CRX242

7. 100 ml of 0.4 M -acidified $KMnO_4$ solution may be decolourised completely by

(A) 200 ml 1N - $K_2Cr_2O_7$ solution (B) 300 ml 0.5M - H_2O_2 solution

(C) 100 ml 0.8N - KI solution (D) 75 ml 1.4 N - $H_2C_2O_4$ solution

CRX064

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8. The correct information(s) about the n-factor of given species in the reaction :
$KMnO_4 + H_2SO_4 + HCl \rightarrow K_2SO_4 + MnSO_4 + Cl_2 + H_2O$ is/are -

(A) $KMnO_4 \rightarrow 5$ (B) $H_2SO_4 \rightarrow 10/3$

(C) $HCl \rightarrow 1$ (D) $K_2SO_4 \rightarrow 2$

CRX367

9. 0.2 mole of Cl_2 is reacted with 500 ml of 1.2 M - NaOH solution

$Cl_2 + OH^- \rightarrow ClO_3^- + Cl^- + H_2O$

The incorrect information(s) related is/are -

(A) Cl_2 is limiting reagent (B) 0.4 mole NaOH will left unreacted

(C) $[ClO_3^-]_{\text{final solution}} = 0.133$ M (D) $[Cl^-]_{\text{final solution}} = 0.267$ M

CRX368

MATRIX MATCH TYPE QUESTION

Oxidation Number

10. Match List-I (Compounds) with List-II (Oxidation states of Nitrogen) and select answer using the codes given below the lists :

List-I	**List-II**
(a) NaN_3	(1) +5
(b) N_2H_2	(2) +2
(c) NO	(3) –1/3
(d) N_2O_5	(4) –1

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(A) (a)-3; (b)-4; (c)-2; (d)-1

(B) (a)-4; (b)-3; (c)-2; (d)-1

(C) (a)-3; (b)-4; (c)-1; (d)-2

(D) (a)-4; (b)-3; (c)-1; (d)-2

CRX108

11. Match the column-

Column-I	**Column-II**
(A) $\underline{P_2H_4} \longrightarrow PH_3 + P_4H_2$	(p) $E = \frac{3M}{4}$
(B) $\underline{I_2} \longrightarrow I^{\ominus} + IO_3^{\ominus}$	(q) $E = \frac{3M}{5}$
(C) $MnO_4^{\ominus} + \underline{Mn^{2\oplus}} + H_2O \longrightarrow Mn_3O_4 + H^{\oplus}$	(r) $E = \frac{15M}{26}$
(D) $\underline{H_3PO_2} \longrightarrow PH_3 + H_3PO_3$	(s) $E = \frac{5M}{6}$

(A) A-r; B-q; C-r; d-s (B) A-s; B-q; C-r; d-p

(C) A-p; B-q; C-r; d-s (D) A-s; B-p; C-q; d-r

CRX068

For More Join: @IITJEE_Advanced

ASSERTION REASON TYPE QUESTION

12. **Statement–1 :-** Moles of $KMnO_4$ required for oxidation of Fe^{2+} in acidic and basic medium will be different.

Statement–2 :- Final oxidation state to which Mn^{7+} will be reduced will be different in case of acidic and basic medium.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

CRX369

13. **Statement–1 :-** Mass of a particular substance that combine with 8 gm of oxygen is said to be equivalent weight of substance.

Statement–2 :-x gm of metal gave y gm of its oxide, so equivalent weight of metal is $\left(\frac{x}{y-x}\right)\times 8$

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

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CRX370

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Equivalent weight & n-factor

1. What is the equivalent weight of HNO_3 in following reaction ?

$HNO_3 + H_2S \longrightarrow H_2O + NO + S$

CRX269

2. Find out the n_{factor} of IO_3^- in the following disproportionation reaction.

$I_2 \xrightarrow{OH^-} IO_3^- + I^-$

CRX271

Equivalent Concept

3. Metallic tin in the presence of HCl is oxidized by $K_2Cr_2O_7$ to stannic chloride, $SnCl_4$. What volume (in litre) of deci-normal dichromate solution would be reduced by 11.9 gm of tin [Sn = 119]

CRX283

4. Calculate the millimoles of Br_2 produced when 10 ml of 0.1 M BrO_3^- reacts with excess of Br^-.

CRX284

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Redox Titration

5. A 458 g sample containing Mn_3O_4 was dissolved and all manganese was converted to $Mn^{2\oplus}$. In the presence of fluoride ion, $Mn^{2\oplus}$ is titrated with 3 L of $KMnO_4$ solution (which was 1.25 N against oxalate in acidic medium), both reactants being converted to a complex of Mn(III), What was the % of Mn_3O_4 in the sample ?

CRX247

Acid Base Titration

6. How many millimoles of gaseous ammonia will be required to be passed into 30 cm^3 of 1 N H_2SO_4 solution to bring down the acid strength of the latter to 0.2 N ?

CRX244

7. H_3PO_4 is a tri basic acid and one of its salt is NaH_2PO_4. What volume (in ml) of 1 M NaOH solution should be added to 12 g of NaH_2PO_4 to convert it into Na_3PO_4 ?

CRX282

Iodimetric and Iodometric Titration

8. 80 g of a sample of anhydrous $CuSO_4$ was dissolved in water and made to 250 mL. 25 mL of this solution after taking usual precaution was treated with a little excess of KI solution. A white ppt. of CuI and iodine was evolved. The iodine so evolved required 40 mL of 1 M hypo solution. What is the approximate purity of $CuSO_4$ sample ? [Take mol. wt. of $CuSO_4$ = 160 u]

CRX246

For More Join: @IITJEE_Advanced

9. 1 g sample of $AgNO_3$ is dissolved in 50 mL of water. It is titrated with 50 mL of KI solution. The AgI precipitated is filtered off. Excess of KI in filtrate is titrated with M/10 KIO_3 in presence of 6M HCl till all $I^{\ominus}$ converted into ICI. It requires 50 mL of M/10 KIO_3 solution. 20 mL of the same stock solution of KI requires 30 mL of M/10 KIO_3 under similar conditions. Calculate % of $AgNO_3$ in sample. The reaction is :

$KIO_3 + 2KI + 6HCl \longrightarrow 3ICl + 3KCl + 3H_2O$

CRX248

10. What volume (in ml) of 0.5 M-$Na_2S_2O_3$ solution will react completely with 12.7 gm iodine ? (I = 127)

CRX285

Double Titration

11. How many litre of 0.1N HCl are required to react completely with 19 gm mixture of Na_2CO_3 and $NaHCO_3$ containing equimolar amount of these two ?

CRX281

Hardness of water

12. 50 L of a water sample contains 7.3 gm $Mg(HCO_3)_2$. Calculate temporary hardness of the water in terms of ppm of $CaCO_3$.

CRX298

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Bankers Academy JEE

Oxidation and Reduction

13. Classify each of the following unbalanced half–reactions as either an oxidation or a reduction :

(a) $O_2(g) \longrightarrow OH^-(aq)$ (b) $H_2O_2(aq) \longrightarrow O_2(g)$

(c) $MnO_4^-(aq) \longrightarrow MnO_4^{2-}(aq)$ (d) $CH_3OH(aq) \longrightarrow CH_2O(aq)$

(e) $MnO_2(s) \longrightarrow Mn_3O_4(s)$ (f) $FeO(s) \longrightarrow Fe_3O_4(s)$

(g) $CH_3COOH(aq.) \longrightarrow CH_3CH_2OH(aq)$

CRX266

2. Oxidation Number

14. Calculate oxidation number of the underlined elements in the following compounds or ions

(a) $H_4\underline{P}_2O_7$ (b) $\underline{S}_2O_3^{2-}$ (c) $Ca\underline{O}_2$ (d) $\underline{Zn}O_2^{2-}$

(e) $\underline{Cr}O_2Cl_2$ (f) $Ca(\underline{Cl}O_2)_2$ (g) $\underline{C}_3O_2$ (h) $K\underline{I}_3$

(i) $\underline{Fe}_3O_4$ (j) $\underline{Br}_3O_8$ (k) $CaO\underline{Cl}_2$ (l) $\underline{O}_2^+$

(m) $\underline{O}_2^-$ (n) $\underline{C}N^-$

CRX264

15. Calculate oxidation number of the underlined elements in the following compounds or ions

(a) $K[\underline{Co}\ (C_2O_4)_2.(NH_3)_2]$ (b) $K\underline{Al}(\underline{S}O_4)_2.12H_2O$

(c) $[\underline{Fe}(CN)_6]^{4-}$ (d) $\underline{Fe}_{0.93}O$

(e) $H_2\underline{S}_2O_8$ (f) $\underline{Cr}O_{\underline{5}}$

(g) $H_2\underline{S}O_5$

CRX265

For More Join: @IITJEE_Advanced

16. Identify the oxidant (oxidising agent) and reductant (reducing agent) in the following reactions

(a) $Zn(s) + \frac{1}{2}O_2(g) \longrightarrow ZnO(s)$

(b) $Zn(s) + 2H^+(aq) \longrightarrow Zn^{2+}(aq) + H_2(g)$

(c) $10H^+(aq) + 4Zn(s) + NO_3^-(aq) \longrightarrow 4Zn^{2+}(aq) + NH_4^+(aq) + 3H_2O(l)$

(d) $2H_2S(aq) + SO_2(g) \rightarrow 3S(s) + 2H_2O(l)$

(e) $2MnO_4^-(aq) + 16H^+(aq) + 10\,Cl^-(aq.) \longrightarrow 2Mn^{2+}(aq.) + 8H_2O(l) + 5Cl_2(g)$

(f) $2MnO_4^-(aq) + 6H^+(aq) + 5H_2O_2(aq.) \longrightarrow 2Mn^{2+}(aq.) + 8H_2O(l) + 5O_2(g)$

CRX267

Equivalent weight & n-factor

17. 1.12 litre dry chlorine gas at NTP (1 ATM & 273K) was passed over a heated metal when 5.56 g of chloride of the metal was formed. What is the equivalent weight of the metal ?

CRX204

18. 0.98 g of the metal sulphate was dissolved in water and excess of barium chloride was added. The precipitated barium sulphate weighted 0.95 g. Calculate the equivalent weight of the metal.

CRX205

19. Find the n-factor of underlined species in the following non redox reaction.

(1) $NaOH + \underline{H_3PO_4} \longrightarrow NaH_2PO_4 + H_2O$

(2) $NaOH + \underline{H_2SO_4} \longrightarrow NaHSO_4 + H_2O$

(3) $\underline{Ca(OH)_2} + HCl \longrightarrow Ca(OH)Cl + H_2O$

(4) $\underline{Na_2CO_3} + HCl \longrightarrow NaHCO_3 + NaCl$

(5) $\underline{Na_2CO_3} + HCl \longrightarrow NaCl + H_2O + CO_2$

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CRX268

20. Find out the equivalent weight of the underlined species in the following reaction :

(i) $\underline{ClO_3^-} + Fe^{2+} + H^+ \longrightarrow Cl^- + Fe^{3+} + H_2O$

(ii) $CuO + \underline{NH_3} \longrightarrow Cu + N_2 + H_2O$

CRX270

Balancing of Redox Reactions by Oxidation Number Method

21. Balance the following redox reactions by oxidation number method :

(i) $N_2H_4(l) + ClO_3^{\ominus}(aq.) \longrightarrow NO(g) + Cl^{\ominus}(aq.)$ (Basic medium)

(ii) $Cl_2O_7(g) + H_2O_2(aq.) \longrightarrow ClO_2^{\ominus}(aq.) + O_2(g) + H^{\oplus}(aq.)$ (Acidic medium)

(iii) $Cr_2O_7^{2\ominus} + SO_3^{2\ominus} + H^{\oplus} \longrightarrow Cr^{3\oplus} + SO_4^{2\ominus} + H_2O$ (Acidic medium)

(iv) $MnO_4^{\ominus}(aq.) + Br^{\ominus}(aq.) + H_2O(l) \longrightarrow MnO_2(s) + BrO_3^{\ominus}(aq.) + OH^{\ominus}(aq.)$ (Basic medium)

(v) $H_2S(g) + Cl_2(g) \longrightarrow HCl(aq.) + S(s)$ (Acidic medium)

(vi) $Fe_3O_4(s) + Al(s) \longrightarrow Fe(s) + Al_2O_3(s)$

(vii) $I_2(s) + Na_2S_2O_3(aq.) \longrightarrow NaI(aq.) + Na_2S_4O_6(aq.)$

CRX262

For More Join: @IITJEE_Advanced

Balancing of Redox Reactions by Ion Electron Method

22. Balance following half reaction in acidic medium

(i) $H_2S \rightarrow S$ (ii) $SO_3^{2\ominus} \rightarrow S^{2\ominus}$

(iii) $H_2O_2 \rightarrow O_2$ (iv) $Cl_2 \rightarrow Cl^{\ominus}$

(v) $MnO_4^{\ominus} + H^{\oplus} \rightarrow Mn^{2\oplus}$ (vi) $NH_3 \rightarrow NO$

(vii) $C_2O_4^{2\ominus} \rightarrow CO_2$ (viii) $S_2O_3^{2\ominus} \rightarrow S_4O_6^{2\ominus}$

(ix) $Cr_2O_7^{2\ominus} + H^{\oplus} \rightarrow Cr^{3\oplus}$ (x) $MnO_4^{\ominus} \rightarrow MnO_2$

CRX261

23. Balance the following redox reactions by half-reaction method (Ion electron method):

(i) $Cr_2O_7^{2\ominus}(aq.) + Fe^{2\oplus}(aq.) + H^{\oplus}(aq.) \rightarrow Cr^{3\oplus}(aq.) + Fe^{3\oplus}(aq.) + H_2O(l)$

(ii) $XeO_6^{4\ominus}(aq.) + F^{\ominus}(aq.) + H^{\oplus}(aq.) \rightarrow XeO_3(g) + F_2(g) + H_2O(l)$

(iii) $CuSO_4 + KI \rightarrow Cu_2I_2 + K_2SO_4 + I_2$

(iv) $NO_3^{\ominus}(aq.) + Mg(s) + H_2O(l) \rightarrow Mg(OH)_2(s) + OH^{\ominus}(aq.) + NH_3(g)$

(v) $H_3PO_2(aq.) + AgNO_3(aq.) + H_2O(l) \rightarrow H_3PO_4(aq.) + Ag(s) + HNO_3(aq.)$

(vi) $Sn^{2\oplus} + Fe^{3\oplus} \rightarrow Sn^{4\oplus} + Fe^{2\oplus}$

(vii) $H_2SO_3(aq.) + Sn^{4\oplus}(aq.) + H_2O(l) \rightarrow Sn^{2\oplus}(aq.) + HSO_4^{\ominus} + H^{\oplus}(aq.)$

CRX263

Acid Base Titration

24. Calculate volume (in L) of 0.4 M NaOH required to react with following mixture
HCl(1 mol) + H_2SO_4 (2 mol)

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CRX280

25. 4.00 gm NaOH is dissolved in 100 ml HCl solution and the solution is diluted to 500 ml. If 25 ml of the diluted solution exactly requires 10 ml of 0.02 M -Na_2CO_3 solution for titration, calculate the concentration of original HCl solution (in gm/litre).

CRX288

Iodimetric and Iodometric Titration

26. 5 g of pyrolusite (impure MnO_2) were heated with conc. HCl and Cl_2 evolved was passed through excess of KI solution. The iodine liberated required 40 mL of N/10 hypo solution. Find the % of MnO_2 in the pyrolusite.

CRX245

27. 25 ml of 0.4 M -acidified $KMnO_4$ solution is treated with excess of KI solution. The liberated I_2 exactly requires 20 ml of $Na_2S_2O_3$ solution for titration. What is the molarity of $Na_2S_2O_3$ solution ?

CRX287

Back Titration

28. 20 ml of 2M -H_2SO_4 solution is mixed with 30 ml of 3M - NaOH solution. Calculate the volume (in ml) of 4M-HCl solution needed to neutralise the resulting solution.

CRX286

Hardness of water

29. A sample of hard water contains 324 ppm of $Ca(HCO_3)_2$. How many gm of CaO is needed to remove all bicarbonates from 100 L hard water ?

CRX297

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. From the given list, the number of compounds with + 4 oxidation state of Sulphur _____.
SO_3, H_2SO_3, $SOCl_2$, SF_4, $BaSO_4$, $H_2S_2O_7$ **[JEE (Main) 2024]**

CRX371

2. Which of the following cannot function as an oxidising agent? **[JEE (Main) 2024]**

(1) N^{3-} (2) SO_4^{2-} (3) BrO_3^- (4) MnO_4^-

CRX372

3. Match the **List-I** with **List-II**

	List-I (Redox Reaction)		**List-II (Type of Redox Reaction)**
A	$CH_{4(g)} + 2O_{2(g)} \xrightarrow{\Delta} CO_{2(g)} + 2H_2O_{(l)}$	(I)	Disproportionation reaction
B	$2NaH_{(s)} \xrightarrow{\Delta} 2Na_{(s)} + H_{2(g)}$	(II)	Combination reaction
C	$V_2O_{5(s)} + 5Ca_{(s)} \xrightarrow{\Delta} 2V_{(s)} + 5CaO_{(s)}$	(III)	Decomposition reaction
D	$2H_2O_{2(aq)} \xrightarrow{\Delta} 2H_2O_{(l)} + O_{2(g)}$	(IV)	Displacement reaction

Choose the ***correct*** answer from the options given below : **[JEE (Main) 2025]**

(1) A-II, B-III, C-IV, D-I (2) A-II, B-III, C-I, D-IV

(3) A-III, B-IV, C-I, D-II (4) A-IV, B-I, C-II, D-III

CRX383

4. Which of the following reactions are disproportionation reactions?

(A) $Cu^+ \rightarrow Cu^{2+} + Cu$

(B) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

(C) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$

(D) $2MnO_4^- + 3Mn^{2+} + 2H_2O \rightarrow 5MnO_2 + 4H^+$

Choose the correct answer from the options given below: **[JEE (Main) 2024]**

(1) (A), (B) (2) (B), (C), (D) (3) (A), (B), (C) (4) (A), (D)

CRX373

5. Which of the given reactions is not an example of disproportionation reaction ?

[JEE (Main) 2022]

(1) $2H_2O_2 \rightarrow 2H_2O + O_2$ (2) $2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

(3) $MnO_4^{\ominus} + 4H^{\oplus} + 3e^{\ominus} \rightarrow MnO_2 + 2H_2O$ (4) $3MnO_4^{2\ominus} + 4H^{\oplus} \rightarrow 2MnO_4^{\ominus} + MnO_2 + 2H_2O$

CRX049

6. An example of a disproportionation reaction is : **[JEE (Main) 2019]**

(1) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$

(2) $2MnO_4^{\ominus} + 10I^{\ominus} + 16H^{\oplus} \rightarrow 2Mn^{2\oplus} + 5I_2 + 8H_2O$

(3) $2CuBr \rightarrow CuBr_2 + Cu$

(4) $2NaBr + Cl_2 \rightarrow 2NaCl + Br_2$

CRX035

7. 1 mole of PbS is oxidised by "X" moles of O_3 to get "Y" moles of O_2. X + Y = _____

[JEE (Main) 2024]

CRX374

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8. Number of moles of H^+ ions required by 1 mole of MnO_4^- to oxidise oxalate ion to CO_2 is ___.

[JEE (Main) 2024]

CRX375

9. Chlorine undergoes disproportionation in alkaline medium as shown below :

$aCl_2(g) + bOH^-(aq) \longrightarrow cClO^-(aq) + dCl^-(aq) + e\,H_2O(l)$

The values of a, b, c and d in a balanced redox reaction are respectively : [JEE (Main) 2024]

(1) 1, 2, 1 and 1 (2) 2, 2, 1 and 3

(3) 3, 4, 4 and 2 (4) 2, 4, 1 and 3

CRX376

10. $2MnO_4^- + bI^- + cH_2O \rightarrow xI_2 + yMnO_2 + zOH^-$

If the above equation is balanced with integer coefficients, the value of z is _____

[JEE (Main) 2024]

CRX377

11. Given below are two statements :

Statement I: S_8 solid undergoes disproportionation reaction under alkaline conditions to form S^{2-} and $S_2O_3^{2-}$

Statement II: ClO_4^- can undergo disproportionation reaction under acidic condition.

In the light of the above statements, choose the most appropriate answer from the options given below : [JEE (Main) 2024]

(1) Statement I is correct but statement II is incorrect.

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

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CRX378

12. In acidic medium, $K_2Cr_2O_7$ shows oxidising action as represented in the half reaction

$Cr_2O_7^{2-} + XH^+ + Ye^- \rightarrow 2A + ZH_2O$

X, Y, Z and A are respectively are : [JEE (Main) 2024]

(1) 8, 6, 4 and Cr_2O_3 (2) 14, 7, 6 and Cr^{3+}

(3) 8, 4, 6 and Cr_2O_3 (4) 14, 6, 7 and Cr^{3+}

CRX379

13. Some CO_2 gas was kept in a sealed container at a pressure of 1 atm and at 273 K. This entire amount of CO_2 gas was later passed through an aqueous solution of $Ca(OH)_2$. The excess unreacted $Ca(OH)_2$ was later neutralized with 0.1 M of 40 mL HCl. If the volume of the sealed container of CO_2 was x, then x is ____ cm^3 (nearest integer).

[Given : The entire amount of $CO_2(g)$ reacted with exactly half the initial amount of $Ca(OH)_2$ present in the aqueous solution.] [JEE (Main) 2025]

CRX384

14. A 100 mL solution was made by adding 1.43 g of $Na_2CO_3 \cdot xH_2O$. The normality of the solution is 0.1 N. The value of x is _____.

(The atomic mass of Na is 23g/mol) :- [JEE (Main) 2020]

CRX039

15. In order to oxidise a mixture one mole of each of FeC_2O_4, $Fe_2(C_2O_4)_3$, $FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium, the number of moles of $KMnO_4$ required is - [JEE (Main) 2019]

(1) 3 (2) 2 (3) 1 (4) 1.5

CRX037

For More Join: @IITJEE_Advanced

16. The strength of an aqueous NaOH solution is most accurately determined by titrating :
(Note : consider that an appropriate indicator is used) **[JEE (Main) 2020]**
(1) Aq. NaOH in a volumetric flask and concentrated H_2SO_4 in a conical flask
(2) Aq. NaOH in a pipette and aqueous oxalic acid in a burette
(3) Aq. NaOH in a burette and concentrated H_2SO_4 in a conical flask
(4) Aq. NaOH in a burette and aqueous oxalic acid in a conical flask

CRX038

17. Only 2 mL of $KMnO_4$ solution of unknown molarity is required to reach the end point of a titration of 20 mL of oxalic acid (2 M) in acidic medium. The molarity of $KMnO_4$ solution should be _____ M.
[JEE (Main) 2024]

CRX380

18. If 50 mL of 0.5 M oxalic acid is required to neutralise 25 mL of NaOH solution, the amount of NaOH in 50 mL of given NaOH solution is ____g.
[JEE (Main) 2024]

CRX381

19. The density of a monobasic strong acid (Molar mass 24.2 g mol) is 1.21 kg L. The volume of its solution required for the complete neutralization of 25 mL of 0.24 M NaOH is_____$\times 10^{-2}$ mL
(Nearest integer) **[JEE (Main) 2023]**

CRX309

20. 20 mL of calcium hydroxide was consumed when it was reacted with 10 mL of unknown solution of H_2SO_4. Also 20 mL standard solution of 0.5 M HCl containing 2 drops of phenolphthalein was titrated with calcium hydroxide the mixture showed pink colour when burette displayed the value of 35.5 mL whereas the burette showed 25.5 mL initially. The concentration of H_2SO_4 is _____ M
(Nearest integer) **[JEE (Main) 2023]**

CRX310

21. 50 mL of 0.5 M oxalic acid is needed to neutralize 25 mL of sodium hydroxide solution. The amount of NaOH in 50 mL of the given sodium hydroxide solution is : **[JEE (Main) 2019]**
(1) 4 g (2) 2 g (3) 8 g (4) 1 g

CRX036

22. 0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solution. According to equation

$$5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$$

Identify the **correct** statements :
(A) 200 mL of KI solution reacts with 0.004 mol of KIO_3
(B) 200 mL of KI solution reacts with 0.006 mol of H_2SO_4
(C) 0.5 L of KI solution produced 0.005 mol of I_2
(D) Equivalent weight of KIO_3 is equal to $\left(\frac{\text{Molecular weight}}{5}\right)$

Choose the **correct** answer from the options given below: **[JEE (Main) 2025]**
(1) (A) and (D) only (2) (B) and (C) only (3) (A) and (B) only (4) (C) and (D) only

CRX385

23. Thiosulphate reacts differently with iodine and bromine in the reaction given below :

$$2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$$

$$S_2O_3^{2-} + 5Br_2 + 5H_2O \rightarrow 2SO_4^{2-} + 4Br^- + 10H^+$$

Which of the following statement justifies the above dual behaviour of thiosulphate?
(1) Bromine undergoes oxidation and iodine undergoes reduction by iodine in these reactions
(2) Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reaction
(3) Bromine is a stronger oxidant than iodine
(4) Bromine is a weaker oxidant than iodine **[JEE (Main) 2024]**

CRX382

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. The order of the oxidation state of the phosphorus atom in H_3PO_2, H_3PO_4, H_3PO_3 and $H_4P_2O_6$ is

[JEE (Advanced) 2017]

(A) $H_3PO_4 > H_4P_2O_6 > H_3PO_3 > H_3PO_2$

(B) $H_3PO_3 > H_3PO_2 > H_3PO_4 > H_4P_2O_6$

(C) $H_3PO_2 > H_3PO_3 > H_4P_2O_6 > H_3PO_4$

(D) $H_3PO_4 > H_3PO_2 > H_3PO_3 > H_4P_2O_6$

CRX076

2. Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen- **[JEE (Advanced) 2012]**

(A) HNO_3, NO, NH_4Cl, N_2 (B) HNO_3, NO, N_2, NH_4Cl

(C) HNO_3, NH_4Cl, NO, N_2 (D) NO, HNO_3, NH_4Cl, N_2

CRX071

3. In neutral or faintly alkaline solution, 8 moles permanganate anion quantitatively oxidize thiosulphate anions to produce X moles of a sulphur containing product. the magnitude of X is

[JEE (Advanced) 2016]

CRX075

4. For the reaction

$I^{\ominus} + ClO_3^{\ominus} + H_2SO_4 \rightarrow Cl^{\ominus} + HSO_4^{\ominus} + I_2$

The correct statement(s) in the balanced equation is / are : **[JEE (Advanced) 2014]**

(A) Stoichiometric coefficient of $HSO_4^{\ominus}$ is 6 (B) Iodide is oxidized

(C) Sulphur is reduced (D) H_2O is one of the products

CRX073

5. The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc. HNO_3 to a compound with the highest oxidation state of sulphur is ___

(Given data : Molar mass of water = 18 g mol^{-1}) **[JEE (Advanced) 2019]**

CRX080

6. In the chemical reaction between stoichiometric quantities of $KMnO_4$ and KI in weakly basic solution, what is the number of moles of I_2 released for 4 moles of $KMnO_4$ consumed ?

[JEE (Advanced) 2020]

CRX083

7. H_2S (5 moles) reacts completely with acidified aqueous potassium permanganate solution. In this reaction, the number of moles of water produced is **x**, and the number of moles of electrons involved is **y**. The value of (**x** + **y**) is ___.

[JEE (Advanced) 2023]

CRX313

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Question Stem for Question Nos. 8 and 9

A sample (5.6 g) containing iron is completely dissolved in cold dilute HCl to prepare a 250 mL of solution. Titration of 25.0 mL of this solution requires 12.5 mL of 0.03 M $KMnO_4$ solution to reach the end point. Number of moles of $Fe^{2\oplus}$ present in 250 mL solution is $x \times 10^{-2}$ (consider complete dissolution of $FeCl_2$). The amount of iron present in the sample is y% by weight.

(Assume : $KMnO_4$ reacts only with $Fe^{2\oplus}$ in the solution

Use : Molar mass of iron as 56 g mol^{-1})

8. The value of **x** is _____. **[JEE (Advanced) 2021]**

CRX084

9. The value of **y** is _____. **[JEE (Advanced) 2021]**

CRX085

10. To measure the quantity of $MnCl_2$ dissolved in an aqueous solution, it was completely converted to $KMnO_4$ using the reaction,

$MnCl_2 + K_2S_2O_8 + H_2O \rightarrow KMnO_4 + H_2SO_4 + HCl$ (equation not balanced).

Few drops of concentrated HCl were added to this solution and gently warmed. Further, oxalic acid (225 mg) was added in portions till the colour of the permanganate ion disappeared. The quantity of $MnCl_2$ (in mg) present in the initial solution is ____. **[JEE (Advanced) 2018]**

(Atomic weights in g mol^{-1} : Mn = 55, Cl = 35.5)

CRX079

Pankers Academy JEE

11. 5.00 mL of 0.10 M oxalic acid solution taken in a conical flask is titrated against NaOH from a burette using phenolphthalein indicator. The volume of NaOH required for the appearance of permanent faint pink color is tabulated below for five experiments. What is the concentration, in molarity, of the NaOH solution ? **[JEE (Advanced) 2020]**

Exp. No.	Vol. of NaOH (ml)
1	12.5
2	10.5
3	9
4	9
5	9

CRX082

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	B	D	C	D	D	D	C	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	C	A	C	D	A	A	D	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	D	D	B	D	C	C	B	B	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	C	B	A	D	C	C	B	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	A	C	A	B	A	D	C	D	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	C	A	D	C	D	D	D	A	D	B
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	D	A	B	B	A	C	A	C	B	B
Que.	71									
Ans.	D									

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	BD	BCD	ABC	ACD	ABD	ACD	B	ABC	BD	A
Que.	11	12	13							
Ans.	B	A	B							

Bankers Academy JEE

EXERCISE (S)

1. 21 **2.** 5 **3.** 4 **4.** 3 **5.** 50
6. 24 **7.** 200 **8.** 80 **9.** 85 **10.** 200
11. 3.00 **12.** 100
13. (a) Reduction,(b) Oxidation,(c) Reduction,(d) Oxidation,(e) Reduction,(f) Oxidation,(g) Reduction
14. (a) +5, (b) +2, (c) –1, (d) +2, (e) +6, (f) +3, (g) +4/3, (h) –1/3, (i) +8/3, (j) +16/3, (k) 0, (l) +1/2, (m) –1/2, (n) +2
15. (a) +3 (b) +3, +6 (c) +2 (d) $+\frac{200}{93}$ (e) +6 (f) +6 (g) +6
16. (a) Oxidant : O_2 ; Reductant : Zn (b) Oxidant : H^+ ; Reductant : Zn
(c) Oxidant : NO_3^- ; Reductant : Zn (d) Oxidant : SO_2 ; Reductant : H_2S
(e) Oxidant : MnO_4^- ; Reductant : Cl^- (f) Oxidant : MnO_4^- ; Reductant : H_2O_2
17. 20.1 **18.** 72.61 **19.** (1) 1 (2) 1 (3) 1 (4) 1 (5) 2
20. (i) 13.91 or 13.92, (ii) 5.66 or 5.67 **21.** Theoretical **22.** Theoretical
23. Theoretical **24.** 12.50 **25.** 39.42 **26.** 3.48
27. 2.50 **28.** 2.50 **29.** 11.20

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	1	1	3	3	8	8	1	8
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	45	10	2	4	8	4	12	1
Que.	21	22	23							
Ans.	1	1	3							

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	6	ABD	288	6	18	1.875	18.75	126
Que.	11									
Ans.	0.11									

Rankers Academy JEE

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